Test

1. What is Java ?

Ans :-

- Java is a popular programming language.
- Java was created by James Gosling in 1995.
- The first public version was released in 1996.
- Java is handle by the oracle and more than 3 billion devices run java.
- Java is General Purpose, High level programming language, Static, Compiled, Interpreted, Platform Independent, Functional Programming and Object oriented Programming Language.
- Java is used transforming programming language across multiple platform and different types of devices, by smartphones to smart TV.

2. Java is 100% Object Oriented ?

Ans :-

- No, Java is not 100% object oriented language.
- In java it can support primitive data types like int, byte, long, short, etc., this are not a objects.
- Therefore data types like int, float, double, etc., are not a object oriented
- That's why java is not 100% object oriented.

3. Explain Features of Java ?

Ans :-

- **General Purpose :-**
  - Java is using various application like – Web Application, Banking Application, Standalone application, Complex application, Distributed application, Gaming Application.

- **High Level :-**
  - Java is a easily human readable and get easily understandable that's why java is high level programming language.

- **Static :-**
  - In java static because it can need to datatype in java

- **Compiled :-**
  - Compiler is converted source code into bytecode.
  - Interpreter is converted bytecode into machine code.
  - Interpreter is read byte code line by line and converted into machine code.

- **Platform Independent :-**
  - Java is platform independent because the java bytecode which is run on any other operating system, for ex- UNIX OS, MAC OS , LINUX OS , etc.

- **Functional Programming :-**
  - Functional programming it means the passing one function to another function as an argument.

- **Object Oriented Programming Language :-**
  - In Object oriented there are Inheritance, Polymorphism, Encapsulation and Abstraction this terms are mainly used in the java.
  - That's why java is a object oriented programming language.

4. Explain Conditional and Loop Statement with Syntax ?

Ans :-

Conditional Statements :-

- If i want to execute some code on the basis of some condition in that term we should go for the conditional statements.
- There are Five type of conditional statements is if , if else, else if ladder, nested if else and switch case.

- If Statement :-
  - Synatx –

```
if (condition)
{
        // statement
}
```

- If else Statement :-
  - Synatx –

```
if (condition)
{
        // statement
}
else
{
        // statement
}
```

- Else – if ladder Statement :-
  - Synatx –

```
if (condition)
{
        // statement
}
else if (condition)
{
        // statement
}

else if (condition)
{
        // statement
}

else
{
        // statement
}
```

- Nested if else Statement :-
    - Synayx –

```
if (condition)
{
        if (condition)
        {
                // statement
        }
        else
        {
                // statement
        }
```

```
}
else
{
        // statement
}
```

- Switch Case :-
  - Syntax –

```
Switch(condition)
{
        Case 1: // Block of code
        break;

        Case 2: // Block of code
        break;

        Case 3: // Block of code
        break;

        Case 4: // Block of code
        break;

        default : //Block of code

}
```

Loop Statement :-

- If I want to repeat some code on the basis of some condition when we can go for looping statement.

- **For loop :-**

  Syntax-

```
for(initialization; condition; updation)
{
        // block of code
}
```

- **While loop :-**

  Syntax-

```
// initialization
for( condition;)
{
        // block of code
        updation;
}
```

- **Do While loop :-**

  Syntax –

```
// initialization
do
{
        //block of code
        //updation;
}
while(condition)
```

5. What is class and object ?

Ans :-

- **Class –**
  - Class is a collection of similar objects.
  - Class is a template
  - In class there are no memory allocated.
  - It can be exist without any objects

- **Object –**
  - Object is a instance of class
  - Object is real
  - Each object has its own memory
  - Object can not exist without a class

6. What is datatypes Explain its types ?

Ans :-

- **Datatype :-**
- Type of data is called as a datatype.

Types :-

- **Primitive Datatype –**
- Primitive data types is a immutable.
- In Primitive data there are stored directly in the variable.
- There are memory size is fixed.
- Primitive data types are including – byte, short, int, long, float, double, char and Boolean.

- **Non-Primitive Datatype –**
- Non - Primitive data types is a mutable.
- Non - Primitive data types are stored only reference.
- Non - Primitive are memory size is not fixed
- It can included in String , array etc

7. Difference Between Primitive and Non Primitive Datatypes?

Primitive	**Non Primitive**
Immutable	Mutable
Stored directly in the variable	Stored by reference
byte, short, int, long, float, double, char and Boolean.	String , array etc
Checked by value	Checked by reference
Fixed Size	Dynamic Size
Simple operations	Complex operations

8. What is method explain its types ?

Ans –

- **What is method :-**
- Methods can include for loops, while loops, and any other programming components.
- Methods can manipulate attributes associated with an object.

- There are three main types of methods: interface methods, constructor methods, and implementation methods.

9. Difference between JDK , JRE and JVM?

Ans –

Aspect	**JDK**	**JRE**	**JVM**
Purpose	Used to develop Java applications	Used to run Java applications	Executes Java bytecode
Platform Dependency	Platform-dependent	Platform-dependent	JVM is OS-specific, but bytecode is platform-independent
Includes	JRE + Development tools	JVM + Libraries	Class Loader, JIT Compiler, Garbage Collector
Use Case	Writing and compiling Java code	Running a Java application on a system	Convert bytecode into machine code

10. Explain Variable Types ?

11. What is OOPs? Why it needed in java?

Ans –

- OOP stands for Object-Oriented Programming.
- Classes and objects are the two main aspects of object-oriented programming.
- Object-oriented programming will make code maintenance and extensibility easier.
- There are four piller of OOPs - encapsulation, inheritance, polymorphism, and abstraction

12. Explain Inheritance and its types ?

Ans –

- Java Inheritance is a fundamental concept in OOPs.
- Inheritance in Java is a mechanism in which one object acquires all the properties and behaviors of a parent object.
- There are five types of inheritance: single, multiple, hierarchical, multilevel, and hybrid.

Types –

- **Single Inheritance** - One class inherits from one parent class.
- **Multiple Inheritance** - A class inherits from more than one class
- **Hierarchical Inheritance** - Multiple classes inherit from the same parent class.
- **Multilevel Inheritance** - A class inherits from another class, which itself inherits from another
- **Hybrid Inheritance** - A combination of two or more types of inheritance.